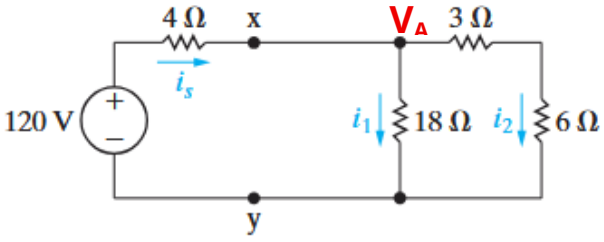
EE 2010
Fall 2026

Name Wyatt Evans

Game 6
9/4/2026

For the following circuit:

a) **Solve for the variables i_s, i_1, and i_2 using Node Voltage**
b) **If we built this circuit in the lab and connected a voltmeter between points x and y, what voltage would we measure?**



(a) Node V_A

$-i_s + i_1 + i_2 = 0$

$i_s = \frac{V_s - V_A}{R_1} = \frac{120 - V_A}{4\Omega}$

$i_2 = \frac{V_A}{R_2} = \frac{V_A}{18\Omega}$

$i_3 = \frac{V_A}{R_3 + R_4} = \frac{V_A}{9\Omega}$

$\frac{120}{4} - \frac{V_A}{4} = \frac{V_A}{18} + \frac{V_A}{9} + \frac{V_A}{4}$

$2 \cdot 36 = \frac{2V_A + 4V_A + 9V_A}{36} = \frac{13V_A}{36}$

$72V = V_A$

$i_s = 12A \quad i_1 = 4A \quad i_2 = 8A$

(b) 72V